INCH-POUND

MIL-E –48491B(AR)
AMENDMENT 2
12 JUNE 1998
SUPERSEDING
AMENDMENT 1
02 MARCH 1988

MILITARY SPECIFICATION

ELECTRONIC UNIT: 11743131

MIL-E-48491 was inactivated after 12 August 1997 for new design.

This amendment forms a part of MIL-E-48491B (AR), dated 30 JUNE 1981, and is approved for use by the US Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agency of the Department of Defense.

PAGE 2

2.1: Add "(REF)" after the following drawing numbers as shown:

"11750202 (REF)
11750439 (REF)
11821471 (REF)
11821520 (REF)
11821748 (REF)
11821457 (REF)
11821458 (REF)"

PAGE 12

3.3.7: Delete in its entirety and substitute the following:

"3.3.7 Four hundred volt supply. With the PFN voltage between J1-A and J1-B at 1200 ± 200 V, the voltage between J2-G and J2-V shall be 390 ± 40 volts. (See 4.5.2.7)"

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* 3.3.10, Line 4: Delete "Figure 4" and substitute "Figure 3".

3.3.11: Line 5: Delete "…the output at J3-g, m, and n in addition to the output pins listed in Table VI shall be 0.4 ± 0.4 vdc and the output at J3-p shall be 1.8 ± 0.8 vdc." and substitute "…the output at J3-b, c, d, q, k, m, and n shall be 0.4 ± 0.4 vdc and the output at J3-I, j, and p shall be 1.8 ± 0.8 vdc.".

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* 3.3.12: Line 4: Delete “Figure 5” and substitute “Figure 4”.

* 3.3.15: Line 2: Delete “J2-i, the connected” and substitute “J2-i, the loads connected”.

3.3.16: Line 2: Delete “applied between J3-x and J3-m” and substitute “applied for not less than 1 sec and not greater than 2 sec between J3-x and J3-m”.

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* Note 1: Delete “Synchronous per Figure 4” and substitute “Synchronous per Figure 3”.

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* TABLE VI, Output condition, column K:

Delete:	and substitute:
“1 ON	1 ON
2 ON	2 ON
3 ON	3 ON
4 OFF	4
5 OFF	5 OFF
6 ON	6 OFF
7”	7 ON”

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* TABLE VIII: Delete in its entirety and substitute the following:

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"TABLE VIII, Target select, $\overline{\text{RESET}}$, $\overline{\text{RBI}}$, $\overline{\text{FEED}}$, and 9995

Item	Input Conditions at Pins of J3 1/, 2/							Outputs 1/, 3/				
	W 7/	AA 7/	Y	Z	A	U	V	J3-m	J3-n	J3-k	Returns 5/	Range 5/ ± 15
1	0	1	1	1	1	0 4/	1	OFF	OFF	ON	0	0000
2	1 4/	0 4/	1	1	1	1	1	OFF	ON	ON	1	830
3	0	1	0	1	1	1	1	OFF	ON	ON	1	830
4	0	1	1	0	1	1	1	OFF	ON	ON	1	830
5	0	1	1	1	0	1	1	OFF	ON	ON	1	830
6	1 4/	0 4/	1	1	1	1	1	OFF	ON	ON	2	1830
7	1 4/	0 4/	1	1	1	1	1	OFF	ON	ON	3	2830
8	1 4/	0 4/	1	1	1	1	1	ON	OFF	ON	4	2830
9	0	1	0	1	1	1	1	ON	OFF	OFF	4	830
10	0	1	1	0	1	1	1	ON	OFF	OFF	4	1830
11	0	1	1	1	0	1	1	ON	OFF	ON	4	2830
12	0	1	1	1	1	1	0	OFF	ON	ON	4	2830
13	0	1	1	1	1	0 4/	1	OFF	OFF	ON	0	0000
14	0	1	1	0	1	1	1	OFF	OFF	OFF	0	0000
15	1 4/	0 4/	1	1	1	1	1	ON	OFF	OFF	1	9995 6/

1/ Input and output signal voltage referenced to Pin J2-v.

2/ Input signal legend: 1 = logical one, digital type – A (item 3.1 of table I).
0 = logical zero, digital type – A signal (item 3.1 of table I).

3/ Output legend: ON = 0.4 ± 0.4 Vdc
OFF = 13.5 ± 2.0 Vdc

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4/ Signal applied for not less than one second and not greater than two seconds.

5/ Readouts to the CLRC.

6/ Range readout may be 000 or 9990.

7/ J3-W is complement of J3-AA."

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* 3.3.17: Delete in its entirety and substitute the following:

"3.3.17 BIT test. With J2-h connected to J2-j, J2-k connected to J2-i, and with a logical one, digital type A signal applied for not less than 1 sec. and not greater than 2 sec. between J6-C and J6-L (return), the output between J6-F and J6-L shall be 0.2 ± 0.2 Vdc, and the readout shall indicate 1830 ± 15 (4.6.2.17)."

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3.5.1: Line 6: Delete "100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec" and substitute "120 ± 12 g's for a duration of 1.7 ± 0.2 milliseconds".

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4.6.1: Delete in its entirety and substitute the following:

"4.6.1 Test equipment. The test equipment in table XIV may be used where specified to perform the required test. Alternate test equipment may be used with the exception of items 3 and 4 of table XIV. If alternate test equipment is to be used, the designs shall be submitted for review and approval to U.S. Army TACOM ARDEC, Picatinny Arsenal, NJ 07806-5000 ATTN: AMSTA-AR-QAC-F."

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4.6.3.1.2: Line 4: Delete "100.0 ± 10.0 g's with a duration of 1.5 ± 0.2 msec" and substitute " 120 ± 12 g's for a duration of 1.7 ± 0.2 milliseconds."

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4.6.3.5: Delete in its entirety and substitute the following:

"4.6.3.5 Temperature Cycling. With power and loads of table I applied and J3-W connected to J3-M, subject the unit to the temperature profile specified in Figure 7. At the conclusion of this test, subject the unit to the performance tests of 4.6.2."

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